Learning Objectives 6.1–6.3: I can calculate the total current through a surface area given a current distribution. I can use Ohm's Law for electromagnetics to calculate the relationship between a current distribution and an electric field. I can articulate and identify what makes a material a good conductor and how conductivity relates to resistance.

Show your work. Write a **Documentation** line at the end (**None** if you did not collaborate). Use $e = 1.602 \times 10^{-19}$ C and $\epsilon_0 = 8.854 \times 10^{-12}$ F/m.

1. **Current through a surface.** A wire of radius $a = 2$ mm lies along the z axis and carries the current density $\mathbf{J} = \hat{\mathbf{z}}\, J_0 \left(1 - \rho^2/a^2\right)$, with $J_0 = 5 \times 10^5$ A/m^2.

   (a) Charge of density ρ_v moves with velocity $\mathbf{u}$. Show that the current through a small area ΔS with unit normal $\hat{\mathbf{n}}$ is $\Delta I = (\rho_v \mathbf{u}) \cdot \hat{\mathbf{n}}\, \Delta S$.

   In a time Δt the charges that cross ΔS are the ones in the tube behind it, of cross section ΔS and length $\mathbf{u}\cdot\hat{\mathbf{n}}\, \Delta t$ along the normal:

$$\Delta Q = \rho_v \left(\mathbf{u}\cdot\hat{\mathbf{n}}\, \Delta t\right) \Delta S$$

$$\Delta I = \frac{\Delta Q}{\Delta t} = (\rho_v \mathbf{u}) \cdot \hat{\mathbf{n}}\, \Delta S$$

   (b) Find the total current in the wire.

   With $d\mathbf{s} = \hat{\mathbf{z}}\, \rho\, d\rho\, d\phi$:

$$I = 2\pi J_0 \int_0^a \left(\rho - \frac{\rho^3}{a^2}\right) d\rho$$

$$= 2\pi J_0 \left(\frac{a^2}{2} - \frac{a^2}{4}\right) = \frac{\pi a^2 J_0}{2}$$

$$= \frac{\pi \times \left(2 \times 10^{-3}\right)^2 \times 5 \times 10^5}{2} = 3.14 \text{ A}$$

   I = 3.14 A

   (c) Find the current inside $\rho < a/2$, and the fraction of the total it carries.

$$I_{1/2} = 2\pi J_0 \left(\frac{(a/2)^2}{2} - \frac{(a/2)^4}{4a^2}\right) = \frac{7}{32}\pi a^2 J_0 = 1.37 \text{ A}$$

$$\frac{I_{1/2}}{I} = \frac{7/32}{1/2} = \frac{7}{16} = 0.44$$

   $I_{1/2}$ = 1.37 A

   fraction = 0.44

   The inner quarter of the area carries 44% of the current, because the current density is largest on the axis.

2. **Ohm's law in point form.** An aluminum wire of cross section 2.5 mm^2 carries 10 A uniformly. Aluminum has $\sigma = 3.7 \times 10^7$ S/m and $n = 1.8 \times 10^{29}$ free electrons per cubic meter.

   (a) Find the current density, the field in the wire, and the voltage across a 20 m length.

$$J = \frac{I}{A} = \frac{10}{2.5 \times 10^{-6}} = 4.0 \times 10^6 \text{ A/m}^2$$
$$E = \frac{J}{\sigma} = \frac{4.0 \times 10^6}{3.7 \times 10^7} = 0.108 \text{ V/m}$$
$$V = E\ell = 0.108 \times 20 = 2.16 \text{ V}$$

   J = 4.0×10^6 A/m^2

   E = 0.108 V/m

   V = 2.16 V

   (b) Find the drift speed of the electrons and their mobility.

$$u = \frac{J}{ne} = \frac{4.0 \times 10^6}{1.8 \times 10^{29} \times 1.602 \times 10^{-19}}$$
$$= 1.39 \times 10^{-4} \text{ m/s}$$
$$\mu_e = \frac{\sigma}{ne} = 1.28 \times 10^{-3} \text{ m}^2/\text{V}\cdot\text{s}$$

   u = 0.139 mm/s

   μ_e = 1.28×10^{-3}

   (c) Aluminum has more free electrons per cubic meter than copper (8.5×10^{28}), yet a lower conductivity. Explain why, using $\sigma = ne\mu_e$.

   The conductivity is the product of the carrier density and the mobility. Aluminum has about twice copper's free-electron density, but its electrons collide with the lattice more often, so their mobility is about a third of copper's (1.28×10^{-3} against 4.3×10^{-3}), and the product is smaller.

   (d) A power line must have the same resistance per meter in aluminum as in copper ($\sigma = 5.8 \times 10^7$ S/m). Find the ratio of the cross sections, and the ratio of the masses per meter. The densities are 2700 and 8960 kg/m^3.

   $R' = 1/(\sigma A)$ is the same, so σA is the same:

$$\frac{A_{\text{Al}}}{A_{\text{Cu}}} = \frac{5.8 \times 10^7}{3.7 \times 10^7} = 1.57$$
$$\frac{m_{\text{Al}}}{m_{\text{Cu}}} = 1.57 \times \frac{2700}{8960} = 0.47$$

   area ratio = 1.57

   mass ratio = 0.47

3. **Resistance of a wire.** A straight wire of length ℓ, cross section A, and conductivity σ has its end at $z = 0$ held at voltage V and its end at $z = \ell$ at zero.

   (a) Assume a uniform field $\mathbf{E} = \hat{\mathbf{z}}\, E$ in the wire. Integrate for the voltage and the current, and derive $R = \ell/(\sigma A)$.

$$V = -\int_{\ell}^{0} E\, dz = E\ell$$
$$I = \int_{S} \sigma E\, \hat{\mathbf{z}}\cdot\hat{\mathbf{z}}\, ds = \sigma E A$$
$$R = \frac{V}{I} = \frac{E\ell}{\sigma E A} = \frac{\ell}{\sigma A}$$

   (b) Explain why the field must have the same magnitude all along the wire when the current is steady.

   A steady current has $\nabla\cdot\mathbf{J} = 0$, so no charge collects between two cross sections and the same current crosses each one. Every cross section has the same area, so J, and $E = J/\sigma$, are the same along the wire.

   (c) Find the DC resistance of 100 m of copper wire of diameter 2.05 mm.

$$A = \pi\left(1.025 \times 10^{-3}\right)^2 = 3.30 \times 10^{-6}\ \text{m}^2$$
$$R = \frac{100}{5.8 \times 10^{7} \times 3.30 \times 10^{-6}} = 0.522\ \Omega$$

$R =$ $0.522\ \Omega$

4. **Leakage through a coaxial line.** A coaxial line has inner radius a, shield radius b, and a dielectric of permittivity ϵ and small conductivity σ.

   (a) Place ρ_ℓ per meter on the inner conductor. Use the field from Gauss's law to derive the leakage conductance per meter, $G' = 2\pi\sigma/\ln(b/a)$.

$$\mathbf{E} = \hat{\boldsymbol{\rho}}\frac{\rho_\ell}{2\pi\epsilon\rho}, \qquad V = -\int_b^a \frac{\rho_\ell}{2\pi\epsilon\rho}\,d\rho = \frac{\rho_\ell}{2\pi\epsilon}\ln\frac{b}{a}$$
$$I = \int_0^\ell\int_0^{2\pi} \sigma\frac{\rho_\ell}{2\pi\epsilon\rho}\,\rho\,d\phi\,dz = \frac{\sigma\rho_\ell\ell}{\epsilon}$$
$$G = \frac{I}{V} = \frac{2\pi\sigma\ell}{\ln(b/a)}, \qquad G' = \frac{G}{\ell} = \frac{2\pi\sigma}{\ln(b/a)}$$

   (b) Moisture raises the conductivity of the dielectric in a 50 m run of RG-58 ($a = 0.405$ mm, $b = 1.475$ mm, $\epsilon_r = 2.25$) to 10^{-12} S/m. Find G' and the DC resistance between the conductors.

G' = 4.86 pS/m

R = 4.1 GΩ

$$G' = \frac{2\pi\times10^{-12}}{\ln(1.475/0.405)} = \frac{6.283\times10^{-12}}{1.293}$$
$$= 4.86\times10^{-12}\ \text{S/m}$$
$$R = \frac{1}{G'\ell} = \frac{1}{4.86\times10^{-12}\times50} = 4.1\ \text{G}\Omega$$

   (c) Lesson 25 gives $C' = 96.8$ pF/m for this cable. Show that $G'/C' = \sigma/\epsilon$ with your numbers.

$$\frac{G'}{C'} = \frac{4.86\times10^{-12}}{96.8\times10^{-12}} = 0.0502\ \text{s}^{-1}$$
$$\frac{\sigma}{\epsilon} = \frac{10^{-12}}{2.25\times8.854\times10^{-12}} = 0.0502\ \text{s}^{-1}$$

The logarithm that carries the geometry cancels in the ratio.

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